

		and then it will accelerate downwards.
3	D	Since trolley A is faster than B, and they have the same mass, the net momentum before collision is towards the right and non-zero. Hence Option A is true. Option B is true by Newton's Third Law. Option C is true. If B is moving to the left then so must be A (as A cannot pass through B) and net momentum will be towards the left which is not possible if momentum is conserved. Option D is false. Whether a collision is elastic or not depends on whether kinetic energy is conserved and not on the duration of the collision.
4	B	Taking moments about the left end of the ruler